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Research article

Dimethyl sulfoxide Toxic and Stimulation Effects in Retinal Pigment Epithelial Cells

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Abstract

Dimethyl sulfoxide (DMSO) is used for drug formulations as a active pharmaceutical ingredients (API) or vehicle of an API. Eye is considered as a relatively isolated organ, protected by barriers similar to the blood-brain one. Thus many systemically administered drugs are not able to reach adequate intraocular levels because they are unable to cross these blood-ocular barriers. Direct injection of drugs into the vitreous cavity has become very popular in recent years. Therefore, it is crucial to establish the level of toxicity and safety of the drugs, since the intraocular tissues are especially sensitive to certain situations that would be perfectly tolerated by other routes of administration. This study has been focused on evaluating the effect on RPE cells directly exposed to DMSO for short to long periods by analyzing cell morphology, cell culture´s confluence level, cell proliferation rate and viability. ARPE-19 cell cultures were exposed to 10%, 5%, 2%, 1%, 0.50%, 0.20% and 0.10% DMSO concentrations and grown for 1, 4 and 14 days. Microscopic observations were performed at 3 hours and 4 days to note any changes in cell morphology and cell culture confluence level. MTT assay was performed to assess cell viability / toxicity and proliferation. The results showed that cells exposed to 10% to 0.50% DMSO were affected for cell morphology and viability / proliferation or only morphology or level of cell culture confluence, even showing dead cell debris as they grew from 1 to 14 days. However, they retained cell morphology, viability and confluent level at concentrations of 0.20% to 0.01% DMSO. But even at lower concentrations, 0.05% to 0.01%, it was noted significant cell proliferation. This study provides data for DMSO effect assessments in RPE cell culture in *in vitro* conditions for measuring cell morphology, viability / toxicity and proliferation, which could be useful in drug formulations. However, this needs to be evaluated in clinical practice as well.

Keywords

Cell toxicity, Cell stimulation, Dimethyl sulfoxide, DMSO, RPE cells

Type of article

Laboratory research with drug formulation relevance

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Introduction

Dimethyl sulfoxide (DMSO; molecular formula (CH3)2SO), an organo-sulfur compound of a relatively high melting point, is a colorless and odorless liquid. DMSO is widely used as a polar aprotic solvent for dissolving nonpolar and polar substances because drugs and chemical substances which are immiscible in water is often soluble in DMSO. DMSO has strong affinity for water on exposure to air which allows diluting it rapidly. It also acts as an oxidant and is reduced to dimethyl sulfide (DMS). It penetrates rapidly as well as allows enhanced penetration of other substances across biological membranes (Capriotti and Capriotti, 2012). This characteristic helps pharmaceutical drugs get through the skin and other body barriers and can affect proteins, carbohydrates, fats, and water in the body. The pharmaceutical industry applies it widely for drug formulations as a vehicle for many active pharmaceutical ingredients (API). Even it is also approved by itself as an API due to its therapeutic properties (Brayton, 1986). It has many therapeutic and toxic effects such as effect on coagulation and anticholinesterase activity (Brayton, 1986). DMSO is also reported to induce histamine release by mast cells, thus regulating inflammation (Brayton, 1986). DMSO is used in palliative treatment for pain, reduces swelling and inflammation in patients with arthritis and connective tissue injuries. It is taken by mouth, topically applied through skin or administered intravenously. The FDA has approved to prescribe it as a palliative treatment for interstitial cystitis (bladder inflammation) and to certain experimental applications (Dinis et al., 2015). A limited number of small clinical trials are reported using intravenous DMSO injection such as experimental treatment of amyloidosis (Swanson, 1985). It is also available on a prescription as a dietary supplement (Brien et al., 2008). Thus, DMSO seems very useful but also controversial due to reported adverse reactions (Jacob and Wood, 1967; Kloverpris et al., 2010; Rubin, 1975; Swanson, 1985; Young et al., 2005). The administrations of safe concentrations of DMSO in combination with drugs are still in question. DMSO probably may interact enhancing toxic effect of some agents used as drugs (Morris et al., 2014; Wood and Wood, 1975; Morris et al., 2014; Ruiz-Delgado et al., 2009).

Eye is considered as a relatively isolated organ, protected by barriers similar to the hemato-encephalic one. This relative physiological advantage becomes a disadvantage at the time of applying treatments, since many of the systemically administered drugs do not reach adequate intraocular levels because they are unable to cross the blood-ocular barriers. This situation has tried to be solved by two approaches: either increases the passage of drugs through the so-called transcorneal route, or injecting the substances directly into the vitreous cavity. The direct injection of drugs into the vitreous cavity has become very popular in recent years, and is used to treat very common diseases such as Age Macular Degeneration (AMD), diabetic macula edema, intraocular infections and many others. However, in these cases it is critical to establish the level of toxicity and safety of the drugs, since the intraocular tissues are especially sensitive to certain situations that would be perfectly tolerated by other routes of administration (Nayak and Misra, 2018).

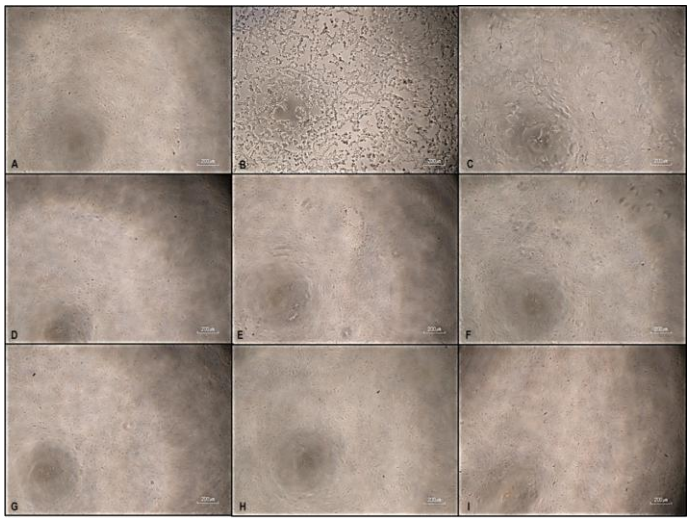


Figure 1. Microscopic observations of ARPE-19 cell cultures exposed to DMSO concentrations and cell culture medium (control). 1 x 10^4 cells per well were seeded in 96 wells culture plats and grown in cell culture medium in standard culture conditions for ≥ 7 days, then exposed to DMSO and cell culture medium. Microscopic observations were performed at 3 hours. Image A represents cell culture medium exposed cell cultures, Images B, C, D, E, F, G, H and I represent to 10%, 5%, 2%, 1%, 0.50%, 0.20% and 0.10% DMSO exposed cell cultures. Images of 0.05%, 0.02% and 0.01% DMSO exposed cell cultures are similar to Images G, H and I, hence they are not presented.

DMSO has been investigated since the 80's as a solvent of different drugs for intraocular use (Kim et al., 2014; Wang et al., 2019; Yoshizumi and Banihashemi, 1988; Yoshizumi et al., 1986). Tolerance of topical DMSO has been tested in several studies; however, topical ocular DMSO used in high concentrations commonly causes ocular irritation (Andrés-Guerrero et al., 2009). Nevertheless, DMSO may enhance some of the effects of several drugs by increasing the speed and volume of tissue penetration, and this made DMSO ideal for getting immediate relief.

Retinal pigment epithelium (RPE) is one of crucial cellular layers of the retina, essential to maintain it structurally and functionally and preventing the onset of several eye diseases. Investigations are going on to find pharmaceutical drugs and their proper safe and effective administrations against eye diseases. Stem cell based products (such as differentiated RPE cell transplantation), also need DMSO for cryopreservation for a long time; therefore, they must be evaluated for the effect of DMSO in each case. Several reports are already published reporting the DMSO dissolved drugs effect over RPE cells. Mostly these reports were made for a specific DMSO concentration exposed to cells for a determined time (Costa de Andrade et al., 2016; Durlu and Tamai, 1997; Madrakhimov et al., 2018; Markowitz et al., 1981). Thus, none of them, to the best of our knowledge, provides an extensive study to evaluate different serial DMSO concentrations along with variable durations of exposures. The study provides data to find how a DMSO concentration could affect RPE cell morphology, their viability and proliferation rates along different exposure times which is essential to know before measuring the increase in cellular permeability to a drug due to DMSO, with the final purpose is establishing safety and efficacy of the drug for patients.

Materials and methods

Cell culture

Human retinal pigment epithelial cell line (ARPE-19) was purchased from American Tissue Culture Collection (ATCC). The cells were grown in DMEM/F12 cell culture medium supplemented with 10% fetal bovine serum (FBS) and 1% antibiotic and antimycotic solution (100 U / ml Penicillin, 100 µg / ml Streptomycin and 0.25 µg / ml of Amphotericin B) (Gibco®, Paisley, UK) in standard culture conditions of 37 oC and 5% CO2. The cell culture medium was replaced every 2-3 days with the fresh medium. When the cell culture reached to 80-90% confluent stage, it was treated with 0.05% Tripsin-EDTA solution (Gibco®, Paisley, UK) and was sub-cultivated.

The cell culture for each experiment execution was prepared in a 96 well culture plate (Costar, Corning Inc., Corning, NY, USA). Single cell suspension of ARPE-19 cells was prepared and cell viability was counted using trypan blue (Sigma-Aldrich®, St. Luis, MO, USA) exclusion method by applying TC20TM automatic cell counter (Bio-Rad Laboratories). The cells showing ~ 90% viability were seeded in cell density of 1 x 10^4 cells per well in 200 µl of cell culture medium using multichannel pipette and grown for 7 days in standard culture conditions and renewing the culture medium every 2-3 days to reach OD570 ≥ 0.2 (ISO 10993-5 standard). Following 7 days growth, the cell culture was synchronized for cell cycle by incubating 24 hours in FBS deprived cell culture medium. At this stage, the ARPE-19 cell cultures were ready for treatments and measuring the effects of treatments.

Cytotoxicity evaluation

Several different concentrations of DMSO (test samples) and cell culture medium (reference sample) were used in this study. The DMSO concentrations were prepared in groups; group 1 (10%, 1% and 0.1%), group 2 (5%, 0.5% and 0.05%), group 3 (2%, 0.2% and 0.02%) and group 4 (1%, 0.1% and 0.01%) by diluting DMSO in cell culture medium. The cell culture medium was removed from each well of cell culture plates following cell cycle synchronization. Then the ARPE-19 cell cultures were incubated with cell culture medium and different DMSO concentrations for 1 to 14 days and by renewing it at every 2-3 days. Since the experiment was performed following ISO guideline 10993-5, therefore, cell cultures growing in the last right and left columns of wells of each cell culture plate were incubated only with cell culture medium. At the end of each exposure period, the cell culture of each well was observed microscopically and any alteration in cell morphology and/or cell culture confluence were noted. At 3 hours and 4 days the cell cultures were photographed. Finally the MTT assay was performed to measure the cytotoxicity produced in cell culture of each well. The spectrophotometer SpectraMax® M5 was used to record the absorbance (optical density; OD) at a wavelength of 570 nm (OD570) and 650 nm (OD650; background absorbance).

Statistical analysis

Data was analyzed following ISO guideline 10993-5. Background absorbance at 650 nm was subtracted from the reading at 570 nm for obtaining final value of the OD570. The formula applied was for measuring the percentage viability was 100 x OD570 (test sample) / OD570 (control sample). The percentage presented in the bar graph is the mean of percentages obtained in each case. Data showed statistically significant difference between one group to another group if p=0.05. Each group was in triplicate (n=3).

Results

Microscopic assessment

Cell morphology and cell culture confluence level

3 hours assessment

Cell culture of each well was observed under microscope and any alteration in cell morphology and confluent levels of cell culture were noted and photographed (Figure 1). The cell cultures exposed to only cell culture medium were confluent and did not show any alteration in cell morphology. They were elongated to polygonal in shape with regular dimensions (Figure 1 A). The cell cultures exposed to 10% DMSO concentration showed small patches of cells (no confluent) and showed higher numbers of cells in rounded shapes (Figure 1 B). In comparison to cell cultures exposed to 10% DMSO, the cell cultures exposed to 5% DMSO concentration were more confluent (i.e. less patches of cells) and showed higher numbers of cells in elongated to polygonal in shape with more regular dimensions (Figure 1 B).

The degrees of cells arrangements in confluent layers and adaptation to the elongated to polygonal shapes and more regular dimensions were increased serially as the concentrations of DMSO in cell cultures were decreased serially from 5% towards 2%, 1%, 0.50%, 0.20%, 0.10%, 0.05%, 0.02% and 0.01% (Figure 1).

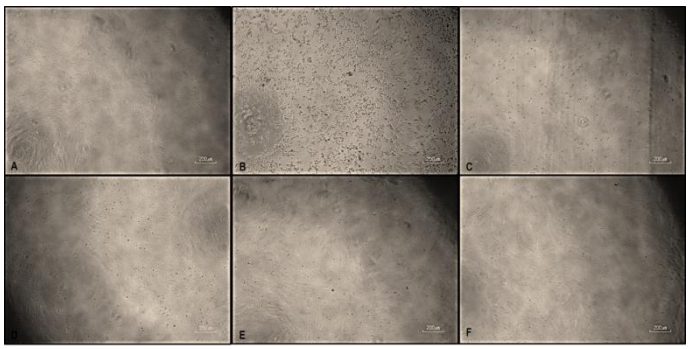


Figure 2. Microscopic observations of ARPE-19 cell cultures exposed to DMSO concentrations and cell culture medium (control). 1 x 10^4 cells per well were seeded in 96 wells culture plate and grown in cell culture medium in standard culture conditions for ≥ 7 days, then exposed to DMSO and cell culture medium. Microscopic observations were performed on the 4th day. Image A represents cell culture medium exposed cell cultures, Images B, C, D, E and F represent to 10%, 5%, 2%, 1% and 0.50%, DMSO exposed cell cultures and dead cell debris. Images of 0.20%, 0.10%, 0.05%, 0.02% and 0.01% DMSO exposed cell cultures are similar to Images F with decreasing or no dead cell debris, hence they are not presented.

4 days assessment

The Cell cultures were also photographed at 4 days. The assessments were done for cells arrangement in a confluent layer and any alteration in cell morphology was noted. The cell cultures exposed to only cell culture medium were highly compacted due to the growing number of cells. Cell morphology was similar as had been seen in images of 3 hours (Figure 2 A). The cell cultures exposed to 10% DMSO contained more separated and rounded cells than the cells arrangements in small patches of cells, which had been seen at 3 hours (Figure 2 A). They were dead cell debris, considered due to their tiny size in comparison to regular cell size. As the concentrations of DMSO in cell cultures were decreased serially from 5% towards 2%, 1%, 0.50%, 0.20%, 0.10%, 0.05%, 0.02% and 0.01%, the numbers of these dead cell debris were decreased serially, and it increased serially the degree of cell arrangements in compact confluent layers and attaining proper cell morphology (Figure 2).

Viability and proliferation

Experiment quality assessment

The value of OD570 in each well containing cell culture was ≥ 0.2. The viability percentages of cell cultures growing cell culture medium in left and right columns of wells of each cell culture plate was ≥ 90%. The difference between cell viability percentage of this left and right columns from their sum was ≤ 15% in each culture plate.

DMSO cytotoxicity assessment

1st day assessment

After 1 day of incubation period, the cell viability of each well was measured using MTT assay. The cell culture incubated only with cell culture medium was set at 100% viable. The cell cultures exposed to 10%, 5%, 2%, 1%, 0.50%, 0.20%, 0.10%, 0.05%, 0.02% and 0.01% serial concentrations of DMSO were compared with this 100% viable cell culture for assessing the DMSO cytotoxicity on 1-day exposure. The results showed (cell culture viability% ± SD%) 53%±0.02%, 84%±0.08%, 102%±0.09%, 79%±0.07%, 91%±0.06%, 89%±0.11%, 81%±0.07%, 80%±0.07%, 83%±0.01% and 87%±0.02% cell culture viability respectively.

4th day assessment

After 4th day of incubation period, cell culture viability was measured, and compared as described in previous paragraph. The results showed (cell culture% ± SD%) 1%±0.00%, 71%±0.04%, 71%±0.00%, 77%±0.09%, 81%±0.11%, 95%±0.10%, 98%±0.02%, 96%±0.11%, 99%±0.14% and 107%±0.04% cell culture viability respectively for exposed serial DMSO concentrations of 10%, 5%, 2%, 1%, 0.50%, 0.20%, 0.10%, 0.05%, 0.02% and 0.01%.

14th day assessment

On 14th day of incubation period, the results showed (cell culture% ± SD%) 1%±0.00%, 20%±0.03%, 39%±0.07%, 73%±0.11%, 71%±0.07%, 85%±0.05%, 90%±0.01%, 96%±0.18%, 99%±0.14% and 107%±0.04% cell culture viability respectively for exposed serial DMSO concentrations of 10%, 5%, 2%, 1%, 0.50%, 0.20%, 0.10%, 0.05%, 0.02% and 0.01%.

Discussion

The DMSO is well documented for its use as a solvent, oxidant and toxicological profile. This favors its use in many applications. In drug formulations it is widely used as a vehicle for API and it itself is evaluated as an API in several orphan-status drugs development (Lutz et al., 2016; Sardana et al., 2011).In research laboratories, it is regularly used for cryopreservation of different types of cells usually for a long period of time in liquid nitrogen. This solvent is capable of stabilizing cell membranes under rapidly changing conditions, preventing intracellular ice crystal formation during freezing and heat release during the period of phase transition. Additionally it is used in the bioassay tests for non or poorly water soluble drugs for different assessments including cytotoxicity (Costa de Andrade et al., 2016; Durlu and Tamai, 1997; Yoshizumi and Banihashemi, 1988; Yoshizumi et al., 1986). Despite of the multiple applications of DMSO, its physiological and pharmacological effects are still not fully understood, that's produced several reports informing the adverse effects in patients. Hence, it becomes important to evaluate if a DMSO concentration itself produces morphological changes, cytotoxic and proliferative effects on cells, specially applying those cells, which are targeted by an API under research and development. Currently there are several eye diseases, which need treatments in a short period due to their big social-economic impacts.

In literature, several types of cells are reported to be tested for assessing DMSO effect. Nevertheless, this study is focused on RPE cells that make a crucial layer of retina and it is well-documented that damage of retina produces vision loss. As said in earlier introductory part, eye is an isolated organ and has barriers blocking the entrance of many drugs systemic administered. Hence, retina and its cellular layers must be tested for DMSO concentrations effects before applying a DMSO based drug formulation as a treatment. In our knowledge, there is no report in which simultaneously in a same experimental design, DMSO is tested for lower to higher concentrations and from a short to a long period for assessing its role in cell morphology changes, cell toxicity production and cell proliferation stimulation. Such research study would definitely support scientific community in deciding DMSO application in their research and effective treatment development purpose.

This research study uses the cells of ARPE-19 cell line, which is well-documented for its application in testing different research approaches at initial laboratory level. Further, in advanced research level, the fresh primary cultures of human RPE cells from donor eyes are recommended before transferring a treatment approach into in vivo tests. Nevertheless, primary human RPE cells shows some limitations such as maintaining protein expressions, trans- differentiation into fibroblasts are among others (Srivastava et al., 2013). The concentrations of DMSO used in this study were from 10% to decreasing onwards i.e. 10%, 5%, 2%, 1%, 0.50%, 0.20%, 0.10%, 0.05%, 0.02% and 0.01%. The 10% is the concentration reported in literature for non-cytotoxic effect on 1hour exposure to human peripheral blood mononuclear cells (PBMC) cultures (Kloverpris et al. 2010). It is also reported that the exposure time to DMSO can be more harmful than concentration itself (Kloverpris et al. 2010). The PBMC treated with 10%

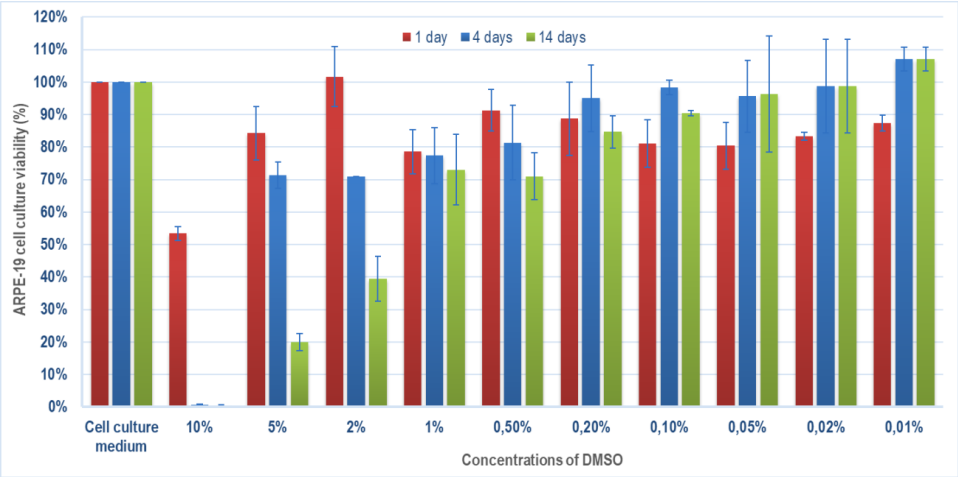


Figure 3 DMSO cytotoxicity assessment in ARPE-19 cell cultures. 1 x 10^4 cells per well were seeded in 96 wells culture plate and grown in cell culture medium in standard culture conditions for ≥ 7 days. Then cell cultures were exposed 10%, 5%, 2%, 1%, 0.50%, 0.20%, 0.10%, 0.05%, 0.02% and 0.01% concentrations of DMSO and cell culture medium as a control for 1, 4 and 14 days. The MTT assay was performed at the end of each exposure periods and data was compared for cell cytotoxicity with cell cultures growing in cell culture medium (100% cell viability). The bar graph showed the results for cell culture% ± SD% vs DMSO concentrations exposed for 1, 4 and 14 days.

DMSO for 1 hour did not affect cells viability, on contrast 0.2% DMSO increased cell death after seven days. Similarly, toxicity was confirmed *in vitro* in a retinal neuronal cell line (RGC-5 cell line) at DMSO concentrations >1% (v/v) (Galvao et al., 2014). There are several such reports in literature providing scientific evidence of executions of different mechanisms producing cell death. It has been reported that high concentrations of DMSO (≥10% v/v) induce toxicity through plasmatic membrane pore formation (de Ménorval et al., 2012; Notman et al., 2006).And there are rare reports that DMSO at low concentration causes toxicity. In EL-4 lymphoma cells, 2.5% (v/v) DMSO induces apoptosis through a mitochondrial-mediated, caspase-3- and caspase-9-dependent pathway (Liu et al., 2001).In RGC-5 cell line, the toxicity produced at lower concentration involves a mechanism based caspase-3 independent neuronal death that involves apoptosis-inducing factor (AIF) translocation from mitochondria to the nucleus and poly-(ADPribose)- polymerase (PARP) activation (Galvao et al., 2014). Nevertheless, the study described in this paper did not focus on exploring the involved mechanisms. But such several reports drive this study to assess the effects of DMSO on cells which depend on the exposure times from a short to long period (1 day to 14 days) along with serially decreasing concentrations. At 14 days, the ARPE-19 cells in a well are totally compact and need frequent nutrient supply i.e. renewal of the culture medium. Therefore, exposure up to 14 days can be considered a good exposure period for reaching a conclusion about the DMSO effect on ARPE-19 cells.

According to ISO guideline 10993-5, a compound is considered non toxic if cell culture viability is ≥ 70%. On the exposure of 10%, the cell cultures retained 53%±0.02% viability at 24 hours which was very close to ≥ 70%. However, cells seemed totally dead at 4 and 14 days. Microscopic observation clearly showed alterations in ARPE-19 cell morphology and cell culture confluent level starting at very early hours i.e. 3 hours. This confirms that exposing the RPE cells to 10% DMSO is extremely not recommended except if the applied therapeutic approach is based on DMSO provoked cytotoxicity. On exposure of DMSO concentrations of 5% to 0.50%, the cell cultures retained average viability was more than 70% at 1 day but at 4 and 14 days, the cell cultures viability was either below to 70% or at margin of 70% or above of 70%. However, the negative value of SD (-SD and +SD) in all these cases were below to 70%. This confirms that 5% to 0.50% DMSO concentrations are non cytotoxic to RPE cells during early hours up to 1 day but they become toxic to cells on long time exposure. Microscopic images taken at 3 hours showed clearly alterations in ARPE-19 cell morphology and cell culture confluent level at 5% DMSO, however, this effect decreased with concentrations along 5% to 0.50% DMSO. Similar was found about the observations of dead cell debris at 4 days which were decreased along 5% to 0.50% DMSO concentrations. Therefore, as previous it is written, the 5% to 0.50% DMSO concentrations is extremely not recommended except if the therapeutic approach is based on DMSO provoked cytotoxicity. The cell cultures exposed to 0.20% to 0.01% DMSO concentrations retained cell cultures viability above to 70%, even the negative value of their SD did not reach to 70%. Microscopic images also did not show any significant alterations in cell morphology and level of cell confluence. This confirms 0.20% to 0.01% DMSO concentrations are non cytotoxic to RPE cells; either cells are exposed for a few hours or long time. It is noted that at concentrations of 0.05% to 0.01% DMSO, cells were proliferating as detected by the MTT assay results at 4 and 14 days, even this proliferation reach above 100% at 0.01% DMSO concentration. This confirms that very low concentrations of DMSO are non cytotoxic to RPE cells but they can stimulate cell proliferation. The study provides useful data for DMSO effect on RPE cells which should be taken into consideration during a drug formulation and non-clinical and clinical experimental studies. However, it should be noted that cell lines usually are more resistant to insults than primary cell cultures, therefore, the effective concentrations of DMSO required for producing an effect (toxicity, proliferation) could be greater in *in vitro* study than *in vivo* one. Thus, this needs to be evaluated in clinical practice as well.

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